Version:1

Bwrdd Iechyd Prifysgol
Betsi Cadwaladr
University Health Board

ANAESTHESIA, CRITICAL CARE & PAIN MANAGEMENT DIRECTORATE

OBSTETRIC ANAESTHESIA WRITTEN CONTROL DOCUMENT

USE OF REMIFENTANIL PCA FOR LABOUR ANALGESIA

Date to be reviewed:	12th March 2029	**No of pages:**	6
Author(s):	Dr Ifan Rees & Dr Gareth Roberts	**Author(s) title:**	Consultant Anaesthetist
Responsible dept / director:	Department of Anaesthesia, BCUHB (West) Dr Lisa Handcock Clinical Lead of Anaesthesia, BCUHB (West)		
Approved by:	Department of Anaesthesia, BCUHB (West)		
Date approved:			
Endorsement by:	Department of Anaesthesia, BCUHB (West)		
Date endorsed:			
Date activated (live):	12th March 2026		

Date EQIA completed:	
Documents to be read alongside this policy:	None
Review	**Purpose of Issue/Description of current changes:** New obstetric anaesthesia written control document for Betsi Cadwaladr University Health Board (West)

First operational:	March 2022				
Previously reviewed:	Date: 1/5/24	Date 11/3/26	Date	Date	date
Changes made yes/no:					

PROPRIETARY INFORMATION
This document contains proprietary information belonging to the Betsi Cadwaladr University Health Board. Do not produce all or any part of this document without written permission from the BCU

Contents

1. Introduction and purpose ……3
2. Scope……..……3
3. Information and safety……...3
4. References……..6

Paper copies of this document should be kept to a minimum and checks made with the electronic version to ensure the version to hand is the most recent.

1. Introduction and purpose

Remifentanil is a safe and effective option for labour analgesia. Acting within 1 - 2 minutes and suited to patient controlled administration, it offers another analgesic option for women in labour. For safe use, the parturient will require careful and continuous monitoring by the midwife during labour.

2. Scope

This guideline is intended as reference for trainee anaesthetists and midwives who wish to provide patient controlled remifentanil analgesia for parturients in labour.

3. Information and safety

3.1 Indications

- If epidural analgesia for labour is contraindicated
- As an alternative to intramuscular (IM) diamorphine for parturients who do not want an epidural
- Labour analgesia following intrauterine death
- Labour analgesia during siting of an epidural
- Analgesia during the repair of tears and episiotomies

3.2 Criteria for use

- Parturient must be over 36 weeks gestation and in active labour
- More than 4 hours must have elapsed since the last dose of IM diamorphine (if administered)
- The Midwifery Shift Leader has confirmed that there is appropriate staffing on labour ward to enable the parturient to have **continuous** trained midwifery presence in the labour room

3.3 Consent

- The parturient will need to have read and understood the patient information sheet given to her by the midwife

- Explain that Remifentanil PCA can be used in conjunction with Entonox

- The anaesthetist should inform the parturient of possible side-effects including drowsiness, itch, nausea and dizziness. In particular that approximately 1 in 10 of parturients using a remifentanil PCA in labour will experience transient lowered SpO_2 levels requiring the administration of oxygen via nasal specs, and that 1 in 100 require the discontinuation of the PCA.

- The parturient should be shown how to use the PCA and should be told to press the button just before or at the start of a contraction.

- Explain that **only** the parturient herself should use the PCA button

3.4 Setting up the PCA

- The anaesthetist on call for the delivery suite is responsible for prescribing and setting up the remifentanil PCA along with the replacing empty pump syringes, while the midwife is responsible for monitoring the patient, the PCA device and ensuring that only the parturient presses the PCA button. A midwife information sheet is also available for their reference.

- Enter the parturient's height at booking (obtained from BadgerNet / handheld notes) into ePMA to calculate the Ideal Body Weight (IBW), which you will require to set up the Remifentanil PCA program on the Arcomed pump.

- Prescribe the Remifentanil PCA order set on ePMA (**including Naloxone**). The PCA prescription reads as :

  Remifentanil PCA (2mg in 40 mls 0.9% Saline (50mcg/ml))

  Bolus dose : 0.5mcg/Kg IBW up to a maximum dose of 40mcg/Kg IBW)

  Lockout period : 2 minutes

- Use a dedicated Arcomed PCA device (kept in labour ward store room, with key for pump attached to the 551 pager)

- Use the Remifentanil PCA Anaesthetic Checklist (attached to the front of the midwife information sheet and observation chart, which is kept in the Remifentanil PCA drawer on labour ward)

- ODP to issue 2mg of remifentanil from the obstetric theatre CD cupboard

- Dilute 2mg of remifentanil with 40ml 0.9% saline in a 50ml syringe to make up a 50 mcg/ml solution

- Label the syringe clearly with the drug name, dilutant and concentration

- Connect the extension tubing onto the syringe, and prime the line
- Load the syringe into the PCA pump
- Ensure the remifentanil profile is selected on the pump (select 'Maternity' – 5511, then 'Remifentanil' – 1115)
- Enter the parturient's Ideal Body Weight (IBW) (as calculated by ePMA) into the pump and check that the bolus dose (0.5mcg/Kg IBW up to max. of 40mcg) and lockout (2 minutes) are correct
- Connect the PCA to a **dedicated** cannula (22g (blue cannula) is sufficient). Remove this cannula **without flushing** when the PCA is no longer needed.
- Ensure that the BP cuff is on the opposite arm to the intravenous cannula used for the PCA, and that no other fluids are run through the same intravenous cannula as the PCA
- Dilute 400mcg naloxone with normal saline to a volume of 4ml in a labelled 5ml syringe (100mcg/ml). This is to be kept by the midwife in the room
- Ensure the oxygen saturation probe has been attached to patient, and that both patient and midwife are happy
- The Remifentanil PCA Anaesthetic Checklist should then be completed
- Start the remifentanil PCA and leave the labour room
- Enter the patient and procedure details in the Obstetric Anaesthesia Procedure and Follow up Book
- The anaesthetist **must remain on labour ward and be immediately available for the first 30 minutes** after starting the PCA. **If the anaesthetist is unable to, the PCA must be stopped**

3.5 Monitoring and Equipment

- **Continuous presence of a trained midwife in the labour room with the parturient at all times** while the remifentanil PCA is being used. If staffing levels or labour ward acuity means this is not possible, **then we cannot offer a remifentanil PCA**.
- **Continuous oxygen saturation monitoring**
- **A bag valve mask should always be available in the room**
- CTG monitoring is not required while the PCA is in progress unless otherwise indicated

- **Documentation of the following observations on the Remifentanil PCA Observation Chart** : Oxygen saturations, respiratory rate, sedation score and pain score, which should be documented every 30 minutes on the Remifentanil PCA Observation Chart.

- **Do not give any other drugs or fluids via the PCA cannula**

- **If oxygen saturations fall < 90%, oxygen via nasal cannula should be commenced at 2L/minute** (1 in 10 women suffer significant respiratory depression with remifentanil). Oxygen saturations < 90% despite nasal oxygen happens in fewer than 1% of women and will require discontinuation of the PCA

3.6 Indications for contacting the anaesthetist

- If the midwife has any concerns

- Sedation score of P or below (AVPU)

- Respiratory rate below 8/min

- Oxygen saturation below 90% despite 2L/min nasal oxygen.

IN THIS SITUATION THE MIDWIFE SHOULD IMMEDIATELY:

- **Stop the remifentanil PCA**

- **Call the anaesthetist to urgently attend**

- **Give a 100mcg dose (1ml bolus of the 100mcg/ml diluted naloxone) every 1 minute until the respiratory rate is > 8/min**

4. References

1. Volikas I, Butwick A, Wilkinson C. Maternal and neonatal side-effects of remifentanil patient-controlled analgesia in labour. BJA 2005; 95: 504